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Article title: From AI Experimentation to Operational Capability: Theoretical Foundations and the Sustained Alignment Framework

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ABSTRACT

We develop the Sustained Alignment Framework (SAF), a governance-centered account of why artificial intelligence initiatives in industrial organizations routinely fail to transition from successful pilots to durable operational capability. Drawing on a structured synthesis of the dynamic capabilities, strategic intelligence, and digital transformation literatures, we argue that this failure is not primarily technical but organizational: analytic capability generates strategic value only when it is interpreted, evaluated, and institutionalized through coherent governance structures. The SAF operationalizes AI-strategy alignment through three interacting dimensions — strategic fit, evaluative coherence, and durable value realization — and positions the Strategic Intelligence Capability Stack (AI-SI-DT) as the conceptual architecture that sustains these dimensions. The present article develops the theoretical foundations of the SAF and describes the archival case-based research design used to investigate alignment mechanisms across large-scale industrial platform enterprises.

Keywords: AI-strategy alignment; sustained alignment framework; strategic intelligence; dynamic capabilities; digital transformation; Industry 4.0; organizational governance

1. INTRODUCTION

A persistent and theoretically significant gap separates the scale of organizational investment in artificial intelligence from the durability of its strategic returns. Across Industry 4.0 environments — defined by cyber-physical systems, platform-mediated coordination, and decentralized data infrastructure — organizations deploy models, accumulate analytic outputs, and expand automation portfolios, yet enterprise-level value remains difficult to sustain beyond the pilot stage (Bharadwaj et al., 2013; Tao et al., 2019; Vial, 2019). The prevailing response to this pattern has been largely technical: improving model accuracy, maturing data pipelines, or accelerating implementation velocity. The present study proceeds from a different premise — that the transition from AI experimentation to operational capability is, in the main, an organizational control problem rather than an engineering one.

Three failure patterns recur with notable consistency across the empirical record. The first, termed use-case opportunism, describes the selection of AI initiatives on the basis of technical feasibility rather than strategic necessity or portfolio coupling. The second, KPI theater, describes the accumulation of performance dashboards that remain decoupled from decision rights and intervention authority, generating visibility without governance consequence. The third, pilot proliferation, describes the persistence of locally successful initiatives outside the operating model routines required to institutionalize them. Taken together, these patterns produce organizations that appear digitally progressive while remaining structurally misaligned.

To address these patterns, we develop the Sustained Alignment Framework (SAF), which operationalizes AI-strategy alignment through three interacting outcome dimensions: strategic fit, evaluative coherence, and durable value realization. The Strategic Intelligence Capability Stack (AI-SI-DT) provides the conceptual architecture through which these dimensions are sustained.

2. THEORETICAL BACKGROUND

The theoretical foundations of the SAF draw on three intersecting bodies of literature: the dynamic capabilities perspective, the strategic intelligence tradition, and the digital transformation literature. Each contributes a distinct analytical layer, and each point toward the same central gap: the absence of a mechanism-based account of the governance and institutional conditions under which AI capability is converted into durable strategic value.

2.1 Artificial Intelligence as Organizational Capability

A growing body of scholarship reconceptualizes artificial intelligence not as a discrete technical artifact but as an organizational capability whose strategic value is produced through the institutional contexts in which it is embedded (Bankins et al., 2024; Wang, 2019). Early adoption-oriented accounts emphasized computational performance, data availability, and implementation readiness as the primary determinants of AI value. Notwithstanding the contributions of this tradition, it systematically underspecifies the governance and evaluative conditions through which technical outputs acquire decision relevance and organizational consequence (Brynjolfsson & McAfee, 2014; Perifanis & Kitsios, 2023). Wang's (2019) Assumption of Insufficient Knowledge and Resources (AIKR) provide an epistemological foundation for a governance-centered account, holding that intelligence must operate under bounded information and constrained resources — implying that evaluative structures, decision rights, and monitoring routines must be designed for uncertainty rather than idealized performance.

2.2 Strategic Intelligence as the Mediating Layer

Strategic intelligence (SI) — defined as the interpretive and governance capability through which organizations transform analytic signals into strategic judgment — constitutes the second theoretical pillar of the SAF (Aguilar, 1967; Day & Schoemaker, 2005; Maccoby, 2015). SI operates through three meta-functional elements: environmental scanning, which detects weak signals and strategic inflection points; strategic interpretation, which translates signals into bounded and actionable meaning; and executive foresight, which converts interpretive commitments into adaptive institutional arrangements (Calof & Wright, 2008). SI prevents AI outputs from accumulating as dashboards and model artifacts that increase informational volume without improving governance judgment.

2.3 Dynamic Capabilities and the Governance of AI

Dynamic capabilities theory offers a complementary lens for understanding AI alignment as an organizational achievement rather than a technical outcome (Teece, 2007; Teece et al., 1997, 2016). Dynamic capabilities — the higher-order abilities to sense environmental change, seize emerging opportunities, and reconfigure assets and processes — provide the organizational conditions under which AI can become strategically consequential. Critically, Teece et al. (2016) locate the challenge of governing AI within the broader problem of asset orchestration in volatile,

uncertain, complex, and ambiguous environments, where the organization's ability to govern AI across its lifecycle constitutes a dynamic capability in itself.

2.4 Digital Transformation as Institutional Embedding

Digital transformation (DT) is theorized as a systemic reconfiguration of organizational logic, structure, and value creation through the integration of digital capabilities (Vial, 2019). Platform architectures, interoperability standards, and workflow integration structures are, in this account, governance decisions that determine whether AI capabilities remain legible, comparable, and governable as the portfolio expands. Matt et al.'s (2015) Four Dimensions of Digital Transformation Strategy underscores that effective transformation requires coherent alignment across technology use, value creation mechanisms, structural changes, and financial architecture — a formulation that anticipates the multi-dimensional alignment construct developed in the SAF.

2.5 A Persistent Theoretical Gap

Taken together, these three bodies of literature converge on a shared insight: the strategic value of AI depends less on computational performance than on the institutional conditions through which that performance is interpreted, evaluated, and governed over time. Notwithstanding this convergence, a theoretically significant gap remains. Existing scholarship provides strong accounts of AI adoption trajectories and digital transformation dynamics, yet gives substantially less systematic attention to the governance and evaluative conditions that allow AI to become a durable organizational capability rather than an isolated technical intervention (Ahmed et al., 2020; Vial, 2019). The SAF addresses that gap by treating alignment not as a milestone achieved but as an ongoing organizational condition.

3. METHODOLOGY

This study employs a qualitative, archival case-based research design appropriate for theory-building purposes (Eisenhardt, 1989; Yin, 2018). The empirical foundation consists of a structured corpus of publicly available organizational documents — including annual reports, corporate governance statements, regulatory filings, platform strategy documents, integrity initiative reports, and disclosed enforcement materials — drawn from large-scale industrial platform enterprises operating in Industry 4.0 contexts. This archival evidence base reflects the logic of documentary analysis and process tracing (Flyvbjerg, 2006): public governance records disclose formal decision

architectures, strategic commitments, evaluative infrastructure, and platform design choices in ways that are stable, auditable, and comparable across organizational settings.

Configurational analysis (Fiss, 2007) provides the overarching analytical strategy. The primary analytic device is the model-based organizational representation: a synthesized configuration constructed from recurring patterns in the archival corpus that specifies how governance, technical, and organizational conditions interact in large-scale industrial platform enterprises. Thematic coding, matrix-based synthesis, clustering, and comparative interpretation organized observed patterns into the configurational findings. It should be noted that many internal governance processes — informal authority patterns, political negotiation, everyday judgment in governance forums — remain only partially visible in public documentation. The present analysis offers a disciplined account of recurrent structural patterns rather than a fully granular reconstruction of organizational interiors.

4. DISCUSSION

The SAF operationalizes sustained alignment through three interacting dimensions, each corresponding to a distinct governance failure mode. Strategic fit refers to the degree to which an AI initiative is positioned as an instrument of enterprise strategy rather than a locally useful technical solution. Fit is operationally present when AI is incorporated into portfolio governance and rendered intelligible within the organization's stated strategic logic; its failure mode is use-case opportunism. Evaluative coherence refers to the degree to which evaluation functions as a governance mechanism rather than symbolic reporting — present when measurement is institutionally tied to decision rights; its failure mode is KPI theater. Durable value realization refers to the degree to which AI becomes an institutionalized operating capability rather than remaining episodic or experimental; its failure mode is pilot proliferation.

Figure 1 illustrates the AI-SI-DT alignment architecture, positioning SI as the mediating layer between AI's computational capacity and DT's structural embedding. In this account, SI prevents AI outputs from accumulating as dashboards and model artifacts that increase informational volume without improving governance judgment: it ensures that predictions, anomaly detections, and algorithmic recommendations enter governance forums as comparable, contestable, and

decision-relevant evidence. Without this mediating function, the risk is not that AI will fail technically but that it will succeed locally while remaining organizationally inert.

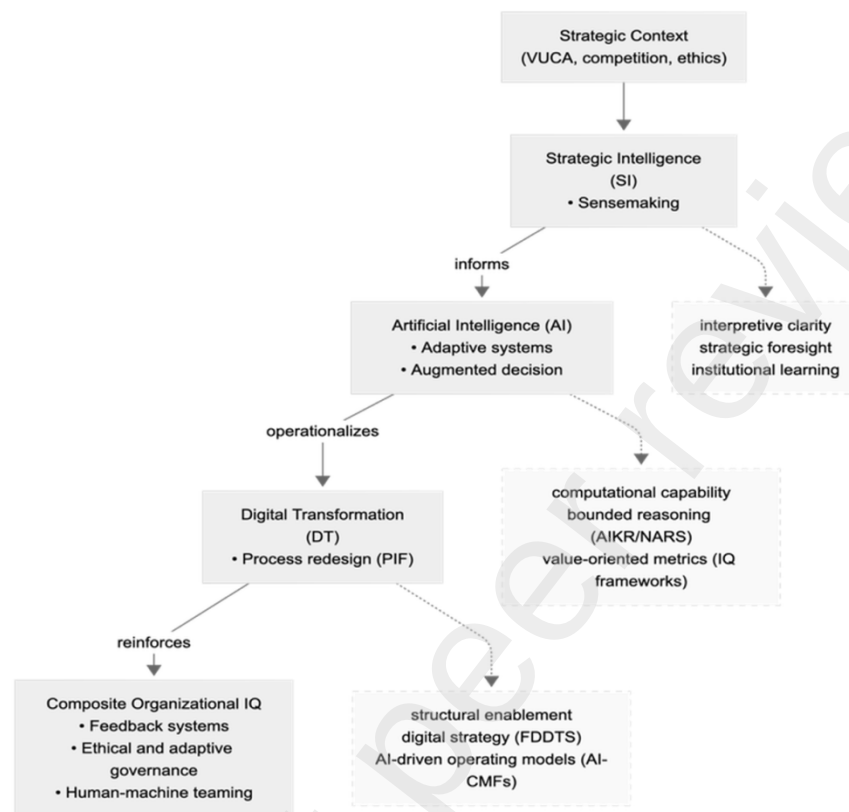


Figure 1. *AI-Mediated Strategic Intelligence and Digital Transformation Alignment. The figure positions SI as the mediating layer linking AI computational capacity to DT structural embedding through interpretive, evaluative, and foresight routines.*

Figure 2 illustrates the Strategic Intelligence Capability Stack, showing how each layer — AI, SI, and DT — contributes to sustained alignment through distinct but mutually conditioning organizational functions. The three SAF dimensions map onto this stack: strategic fit is mediated at the SI layer; evaluative coherence is co-produced at the SI-DT interface; and durable value realization is principally a DT-layer outcome. The stack extends dynamic capabilities theory (Teece, 2007; Teece et al., 2016) by specifying the interpretive and governance mechanisms through which AI-enabled sensing is converted into coordinated organizational response.

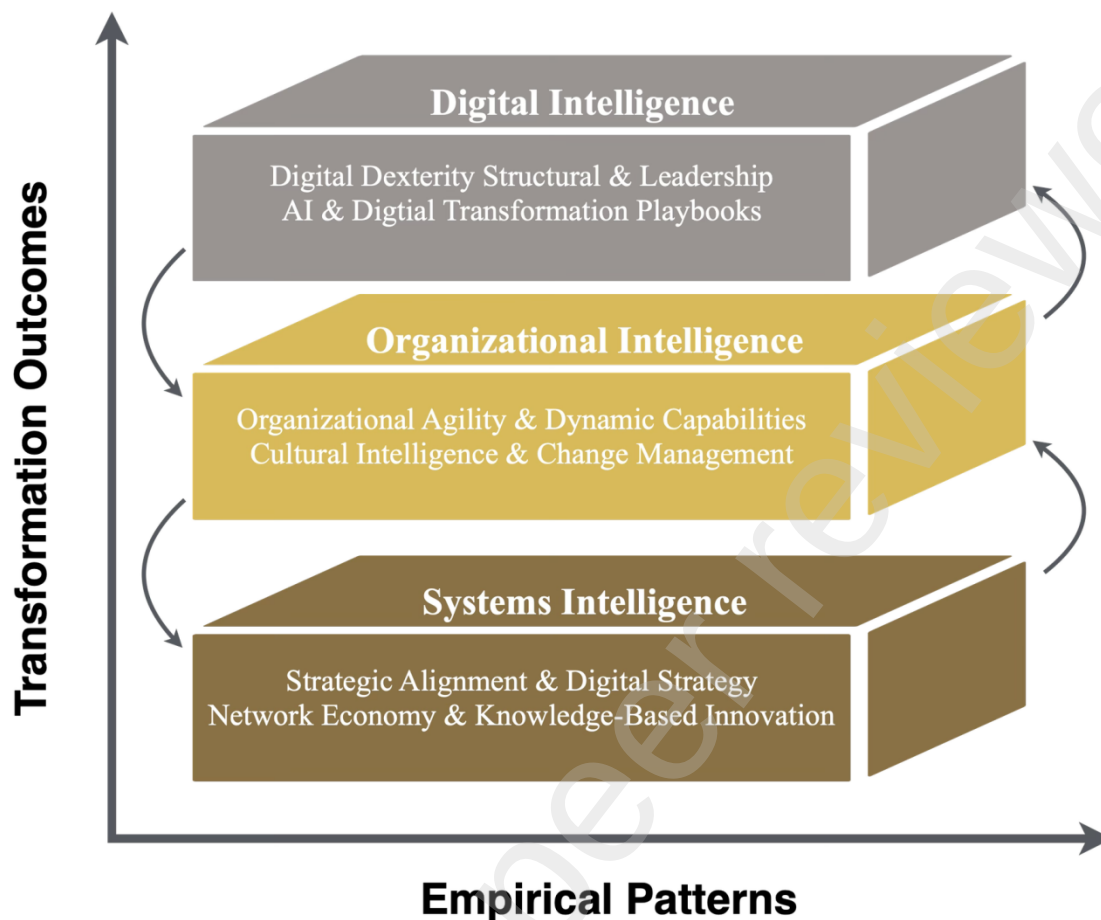


Figure 2. *Strategic Intelligence Capability Stack (AI-SI-DT). The stack illustrates how AI computational capacity, SI interpretive governance, and DT structural embedding function as mutually conditioning layers of organizational intelligence.*

A central analytical implication of the SAF is the importance of capability sequencing: sustained alignment depends on the development of decision rights, governance architecture, and evaluative coherence before — or at minimum alongside — increasing analytic sophistication. Organizations that invert this sequence predictably encounter fragmentation, unmanaged risk exposure, and value dissipation, regardless of the technical quality of their AI systems. This finding extends the evaluative intelligence framework proposed by Liu et al. (2017a, 2017b), which repositions AI assessment from narrow performance measurement toward purpose-driven, multidimensional evaluation explicitly linked to strategic and economic return.

5. LIMITATIONS AND FUTURE LINES OF RESEARCH

This study is not without limitations. The use of publicly available organizational evidence means that many internal governance processes — informal authority patterns, political negotiation, and

everyday judgment in governance forums — remain only partially visible. The SAF is developed through archival configurational analysis of large-scale industrial platform enterprises, and its mechanisms may translate unevenly to small and medium-sized organizations, public sector contexts, or industries with different regulatory and institutional environments. Configurational inference identifies recurring mechanism patterns without establishing universal causal laws; the framework should therefore be read as a theoretically grounded analytical vocabulary rather than a predictive model.

Future research should extend this inquiry through longitudinal designs that trace how alignment strengthens, erodes, or drifts as AI initiatives move from experimentation to institutional embedding — a temporal dimension that archival cross-sectional analysis cannot fully capture. Deeper organizational access, through interview-based or ethnographic methods, would allow scholars to examine how decision rights, escalation routines, and monitoring thresholds operate in practice. The development of explicit operational measures for strategic fit, evaluative coherence, and durable value realization would support systematic empirical testing and refinement of the SAF across diverse organizational settings.

6. IMPLICATIONS FOR THEORY AND PRACTICE

The SAF makes three interconnected theoretical contributions. First, it reframes AI-strategy alignment as a maintained organizational condition rather than an achieved milestone — a framing that has not been proposed with this degree of mechanism-level specificity in prior literature. By operationalizing alignment through three interacting dimensions, the framework enables mechanism-level comparison across organizations and contexts, advancing the theory-building agenda called for by Vial (2019) and Kane et al. (2019). Second, positioning SI as the mediating capability between AI and DT contributes a more organizationally grounded account of AI alignment than technically focused alternatives. Third, the AI-SI-DT stack extends dynamic capabilities theory by specifying the interpretive and governance mechanisms through which AI-enabled sensing is converted into a coordinated organizational response.

For practitioners, the SAF directs attention toward the governance architecture that must accompany — and precede — technical investment. Organizations should not wait for computational maturity before establishing decision rights, evaluative routines, and monitoring

infrastructure. The identification of three failure modes — use-case opportunism, KPI theater, and pilot proliferation — offers practitioners a diagnostic vocabulary for assessing where alignment is breaking down and which governance mechanisms require strengthening. Platform architecture decisions, in particular, function as governance decisions that determine whether AI capabilities remain legible, comparable, and governable as the portfolio expands.

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